

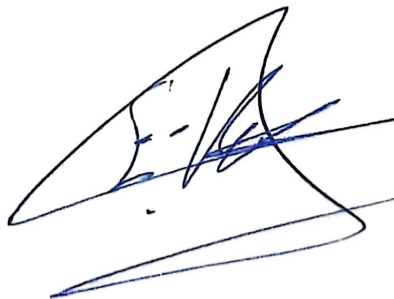
- Derivative -

$$f(x) = \ln \left(\frac{e^{2x} \sqrt{\sin x}}{x^2 + 1} \right)$$

$$\Rightarrow f(x) = [\ln(e^{2x}) + \ln(\sqrt{\sin x})] - \ln(x^2 + 1)$$

$$f'(x) = \left[\frac{2e^{2x}}{e^{2x}} + \frac{\frac{1}{2} \cos x}{\sqrt{\sin x}} \right] - \frac{2x}{x^2 + 1}$$

$$f'(x) = 2 + \frac{\cos x}{2\sin x} - \frac{2x}{x^2 + 1}$$

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